

Ans: Page 11-19

Lecture – 2

8. What is Binary Codes?

Ans: In the coding, when **numbers, letters or words are represented** by a **specific group of symbols**, it is said that the number, letter or word is being **encoded**.

9. Advantages of Binary code.

1. Binary codes are **suitable for the computer applications**.
2. **suitable for the digital communications**.
3. the **analysis and designing of digital circuits much easier**.
4. Since only 0 & 1 are being used, **implementation becomes easy**.

10. Classification of Binary Codes

- Binary Coded Decimal (BCD) Code
- Gray Code
- Excess-3 Code
- Alphanumeric Codes
- Error Detecting Codes
- Error Correcting Codes

11. Explain and Conversion BCD Code.

In this code each **decimal digit is represented by a 4-bit binary number**. BCD is a way to express each of the decimal digits with a binary code. BCD code **only first ten (0000 to 1001)** of these are **used**.

Decimal	0	1	2	3	4	5	6	7	8	9
BCD	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

12. Advantages and Disadvantages of BCD Code.

Advantage:

- It is very **similar to decimal system**.
- We **need to remember binary equivalent** of decimal numbers 0 to 9 only.

Disadvantages:

- The **addition and subtraction** of BCD have different rules
- The BCD **arithmetic** is little more complicated.
- BCD **needs a greater number of bits** than binary to **represent the decimal number**.

13. Define Gray Code.

Ans: The **Gray Code** is a sequence of binary number systems, which is also known as **Reflected Binary Code (RBC)**.

Conversion -> i. Binary to Gray code and ii. Gray to Binary code.

Ans: i. page (3-4) ii. Page (4-5)

14. Explain Gray code characteristics.

- In Gray code, **two consecutive values are differed by one bit** of binary digits
- Binary number is **converted to Gray code to reduce the switching operation**.

15. Conversion Binary to Exces-3 code.

Ans: page (7-8)

➤ Methods for Finding Excess-3 code of Binary Number.

Method 01:

- We **find the decimal number** of the given binary number.
- Then we **add 3 in each digit** of the decimal number.
- Now, we **find the binary code of each digit** of the newly generated decimal number.

Method 02:

- We can also add **0011** in each **4-bit BCD code** of the decimal number for getting excess-3 code.

16. Self-Complementary Property.

Ans: Page (09)

- A self-complementary binary code is a code which is **always complimented in itself**. By replacing the bit **0 to 1** and **1 to 0** of a number, we find the **1's complement** of the number.
- If we perform the **1's complement of excess-3 of a decimal number**, it will be **equal** to the **excess-3 code of the 9's complement of that decimal number**.

17. Difference Between Unicode and ASCII

Unicode Coding Scheme	ASCII Coding Scheme
It uses variable bit encoding according to the requirement. For example, UTF-8, UTF-16, UTF-32	It uses 7-bit encoding. As of now, the extended form uses 8-bit encoding.
It is a standard form	It is not a standard all over the world.
People use this scheme all over the world	It has only limited characters hence, it cannot be used all over the world.
It has more than 128,000 characters.	In contrast, it has only 256 characters

Lecture – 03

18. Define De Morgan's Theorems.

Ans: There are two basic theorems of great importance in Boolean Algebra, which are **De Morgan's First Law** and **De Morgan's Second Law**. These are also called De Morgan's Theorems.

i. De Morgan's First Law: $(P \cdot Q)' = (P)' + (Q)'$

ii. De Morgan's Second Law: $(P + Q)' = (P)' \cdot (Q)'$

19. Applications of Boolean Algebra

- ❖ Digital Logic Design
- ❖ Algorithm Design
- ❖ Telecommunications
- ❖ Artificial Intelligence (AI)
- ❖ Electrical Engineering

The basic laws of Boolean Algebra are summarized in the table below:

Law	OR Form	AND Form
1. Identity Law	$P + 0 = P$	$P \cdot 1 = P$
2. Idempotent Law	$P + P = P$	$P \cdot P = P$
3. Commutative Law	$P + Q = Q + P$	$P \cdot Q = Q \cdot P$
4. Associative Law	$P + (Q + R) = (P + Q) + R$	$P \cdot (Q \cdot R) = (P \cdot Q) \cdot R$
5. Distributive Law	$P + (Q \cdot R) = (P + Q) \cdot (P + R)$	$P \cdot (Q + R) = (P \cdot Q) + (P \cdot R)$
6. Inversion Law	$(A')' = A$	$(A')' = A$

20. Math page (6-10)

Law	OR Form	AND Form
7. De Morgan's Law	$(P + Q)' = P' \cdot Q'$	$(P \cdot Q)' = P' + Q'$
8. Complement Law	$P + P' = 1$	$P \cdot P' = 0$
9. Domination Law	$P + 1 = 1$	$P \cdot 0 = 0$
10. Double Negation Law	$(P')' = P$	$(P')' = P$
11. Absorption Law	$P + (P \cdot Q) = P$	$P \cdot (P + Q) = P$

21. Advantage and Disadvantages of K-Map.

Advantages:

- a) Makes Logic Simpler
- b) Minimizes Logic Gates
- c) Reduce Errors
- d) Time-Saving
- e) No need for Boolean Laws

Disadvantages:

- a) Limited to Fewer Variables
- b) Not suitable for all functions
- c) Space Limitations
- d) Requires Careful Grouping

22. Math page (1-04).

Sequential Circuits

The diagram illustrates the components of a sequential circuit. It features two main blocks: a light blue box labeled "Combinational Circuit" and a light orange box labeled "Memory Elements".

- Inputs:** Two arrows labeled "Inputs" point into the "Combinational Circuit" block from the left.
- Outputs:** An arrow labeled "Outputs" points out of the "Combinational Circuit" block to the right.
- Feedback Loop:** An arrow originates from the bottom of the "Combinational Circuit" block, goes down and then left, and enters the "Memory Elements" block.
- Output from Memory:** An arrow originates from the top of the "Memory Elements" block, goes up and then right, and enters the "Combinational Circuit" block, completing a feedback loop.

Combinational Circuits	Sequential Circuits
1. The outputs of the combinational circuit depend only on the present inputs.	1. The outputs of the sequential circuits depend on both present inputs and present state (previous output).
2. The feedback path is not present in the combinational circuit.	2. The feedback path is present in the sequential circuits
3. In combinational circuits, memory elements are not required.	3. In the sequential circuit, memory elements play an important role and require.
4. The clock signal is not required for combinational circuits.	4. The clock signal is required for sequential circuits.
5. The combinational circuit is simple to design	5. It is not simple to design a sequential circuit.

- ii. Positive level triggering
- iii. Negative level triggering
- iv. Edge triggering
- v. Positive edge triggering

5. Define Latch and explain SR Latch for NAND Gate.

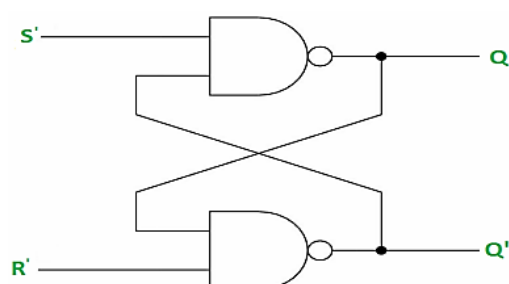
Ans: Latches are basic storage elements that operate with signal levels. Latches controlled by a clock transition are flip-flops. Latches are level-sensitive devices.

SR (Set-Reset) Latch – The SR latch forms the basic building blocks of all other types of flip-flops.

SR Latch is a circuit with:

- (i) 2 cross-coupled NAND gate.
- (ii) 2 input S for SET and R for RESET.
- (iii) 2 output Q, Q'.

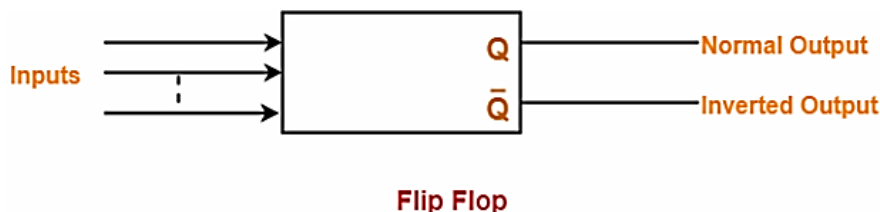
Under normal conditions, both the input remains 0. The following is the RS Latch with NAND gates:



Q	Q'	STATE
1	0	Set
0	1	Reset

6. Define Flip – Flop and type of flip – flop .

Ans: A Flip Flop is a memory element that is capable of storing one bit of information.



Type of Flip-Flop: i) SR Flip Flop ii) JK flip flop iii) D flip flop iv) T flip flop

Logic Symbol-

The logic symbol for SR Flip Flop is as shown below-



7. Explain SR Flip Flop for Nand Latch.

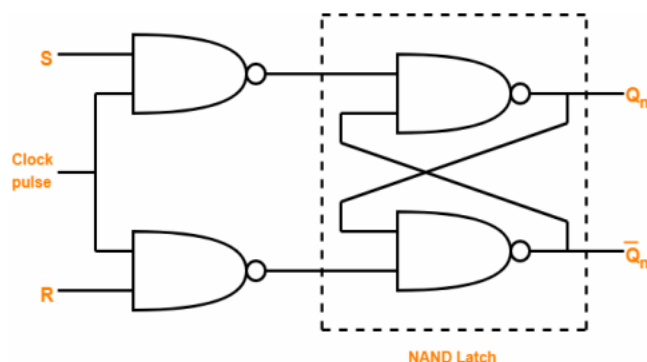
Ans: • SR flip flop is the simplest type of flip flops.

- It stands for **Set Reset flip flop**.
- It is a clocked flip flop

Construction of SR Flip Flop By Using NAND Latch -

- NAND latch
- Two NAND gates

Logic Circuit:



Truth Table-

The truth table for SR Flip Flop is as shown below-

INPUTS			OUTPUTS
S	R	Q _n (Present State)	Q _{n+1} (Next State)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Indeterminate
1	1	1	Indeterminate

8. Define JK Flip Flop.

Ans: JK flip flop is a refined & improved version of **SR Flip Flop** that has been introduced to solve the problem of indeterminate state that occurs in SR flip flop when both the inputs are 1.

9. Explain JK Flip Flop for Nand Latch.

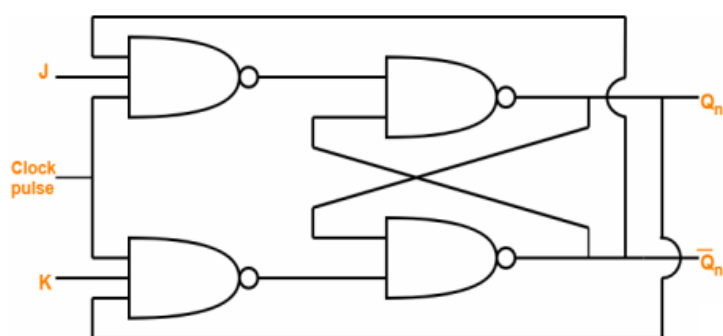
Ans: Construction of JK Flip Flop By Using SR Flip Flop Constructed From NAND Latch –

This method of constructing JK Flip Flop uses-

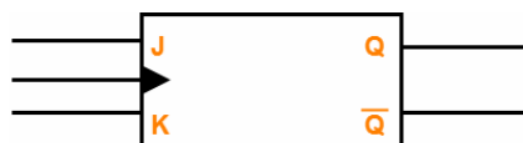
- SR Flip Flop constructed from NAND latch
- Two other connections

Logic Circuit –

The logic circuit for JK Flip Flop constructed using SR Flip Flop constructed from NAND latch is as shown below-



Logic Circuit For JK Flip Flop Using SR Flip Flop
(Constructed Using NAND Latch)



Logic Symbol

Truth Table-

The truth table for JK Flip Flop is as shown below-

INPUTS			OUTPUTS
J	K	Q_n (Present State)	Q_{n+1} (Next State)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

10. SR Flip Flop VS JK Flip Flop.

Ans: Both JK flip flop and SR flip flop are functionally same.

The only difference between them is-

- In JK flip flop, indeterminate state does not occur.
- In JK flip flop, instead of indeterminate state, the present state toggles.
- In other words, the present state gets inverted when both the inputs are 1.

11. Explain D Flip Flop with Circuit diagram and truth table.

The D flip flop is the most important flip flop from other clocked types. It ensures that at the same time, both the inputs, i.e., S and R, are never equal to 1.

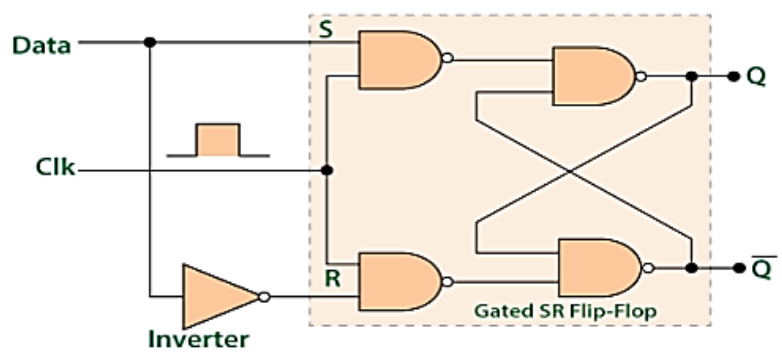
This single data input, which is labeled as "D" used in place of the "Set" input and for the complementary "Reset" input, the inverter is used. Thus, the level sensitive D-type or D flip flop is constructed from a level-sensitive SR flip flop.

So, here $S=D$ and $R=\sim D$ (complement of D)

Block Diagram



Circuit Diagram:



We know that the SR flip-flop requires two inputs, i.e., one to "SET" the output and another to "RESET" the output.

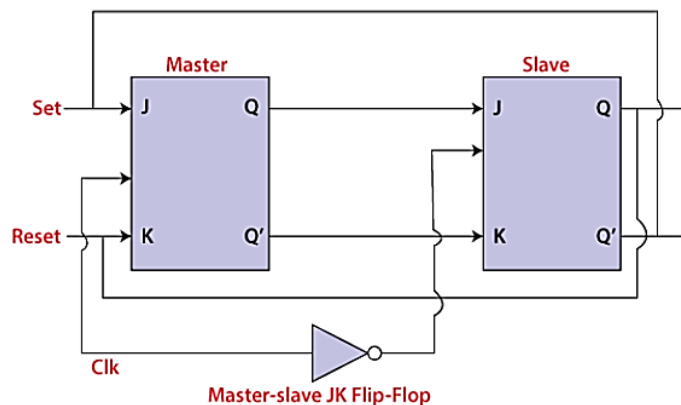
The "CLOCK" or "ENABLE" input is used to avoid this for isolating the data input from the flip flop's latching circuitry. When the clock input is set to 1, the "set" and "reset" inputs of the flip-flop are both set to 1. In simple words, the output is "latched" at either 0 or 1. In simple words, the output is "latched" at either 0 or 1.

Truth Table for the D-type Flip Flop

Clock	D	Q	Q'	Description
↓ » 0	X	Q	Q'	Memory no change
↑ » 1	0	0	1	Reset Q » 0
↑ » 1	1	1	0	Set Q » 1

12. Explain Master – Slave JK Flip Flop with Timing diagram.

The master-slave flip flop is constructed by combining two JK flip flops. These flip flops are connected in a series configuration. In these two flip flops, the 1st flip flop work as "master", called the master flip flop, and the 2nd work as a "slave", called slave flip flop. In "master-slave flip flop", apart from these two flip flops, an inverter or NOT gate is also used.

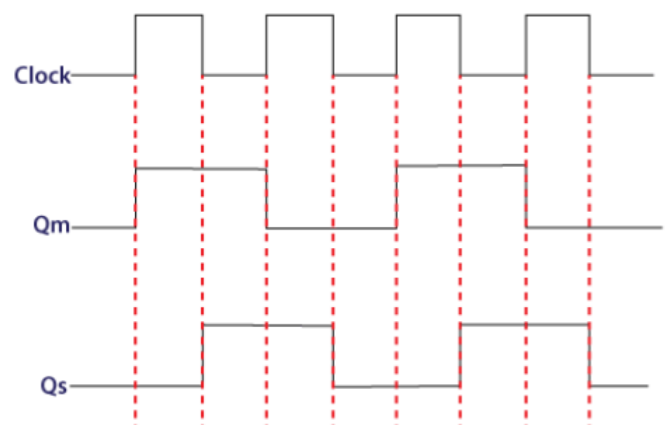


Working process:

1. When **CP = 1**, the master flip-flop responds to the **J and K inputs**, while the slave remains isolated.
2. When **CP changes from 1 to 0**, the master stops responding and the slave receives the master's output.
3. For **J=1, K=0**, the output is set; for **J=0, K=1**, the output is reset; and for **J=K=1**, the output toggles.
4. When **J=K=0**, the flip-flop is disabled, so the output **Q remains unchanged**.

Timing Diagram of a Master Flip Flop:

1. When **CP = 1**, the master flip-flop is active and responds to the input, while the slave flip-flop remains inactive.
2. When **CP = 0**, the master holds its output and the slave flip-flop becomes active.
3. The slave copies the master's output and provides the final output until the clock changes again.



13. Define Counter in Digital Electronics.

Ans: A counter is a digital circuit that can be used to keep track of the number of events or clock cycles that have occurred.

i. Binary Counter ii. Decade/ Decimal counter iii. Up/Down counter

14. Define Register in Digital Electronics.

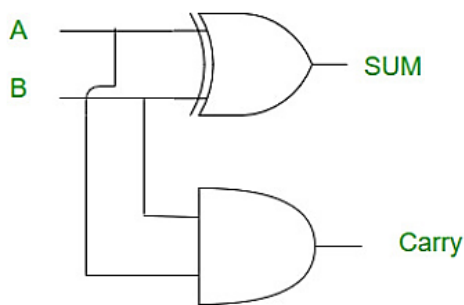
Ans: A register in digital electronics is a group of flip-flops that can be used to store binary data. Registers are often used to hold intermediate results in digital systems

Half, Full Adder, Ring and Johnson Counter

15. Explain Half Adder with diagram.

Ans: A half adder is a basic combinational circuit that adds two single-bit binary inputs (A and B) to produce a SUM using an XOR gate and a CARRY using an AND gate, without considering any carry-in from a previous stage.

- Performs binary addition of two single-bit inputs, generating a SUM ($A \oplus B$) and CARRY ($A \cdot B$)



Truth Table of Half Adder

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

16. Half Adder Advantages and Disadvantages and Application.

Advantages:

- Simple Design
- Fast Operation
- Low Hardware Requirement

Disadvantages:

- No Carry Input
- Limited Functionality
- Requires Full Adders

Application:

- Binary Addition
- Building Block for Full Adders
- Digital Arithmetic Circuits

17. Explain Full Adder with Circuits Diagram.

Ans: Full Adder is a combinational circuit that adds three inputs and produces two outputs. The first two inputs are A and B and the third input is an input carry as C-IN. The output carry is designated as C-OUT and the normal output is designated as S which is SUM.

Logic Circuit of Full Adder

To implement a Full Adder using basic logic gates:

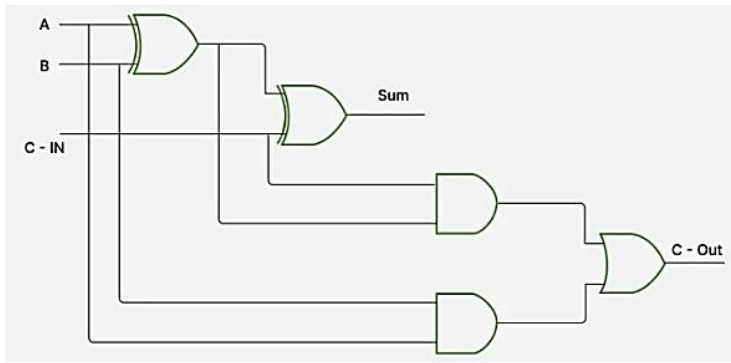
Sum (S) is implemented using XOR gates:

Use two XOR gates:

- First XOR gate: $A \oplus B$

- Second XOR gate: $(A \oplus B) \oplus C\text{-IN}$ to get the final sum S

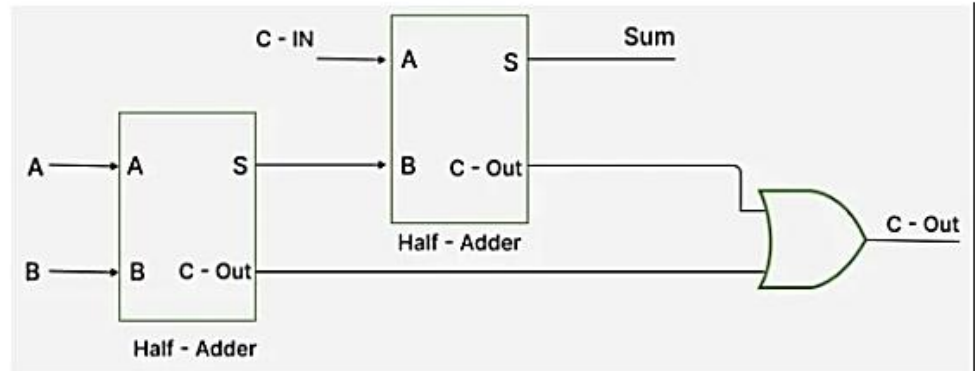
N.B : দুইটা Half Adder দিয়ে একটা Full Adder create করা



Sum: $S = A \oplus B \oplus C\text{-IN}$

$C\text{-OUT} = AB + C\text{-IN}(A \oplus B)$

Implementation of Full Adder using Half Adders



18. Define Ring Counter.

Ans: A ring counter is a typical application of the Shift register. the last flip-flop is connected to the input of the first flipflop in the case of the ring counter.

No. of states in Ring counter = No. of flip-flop used

19. 3, 4, 5 bit Ring Counter Example with diagram and working process.

Ans: Let's consider a 4-bit Ring Counter:

1. Initial state : 1000

2. Clock Cycle 1: The 1 bit shifts to the next flip-flop, resulting in the state 0100.

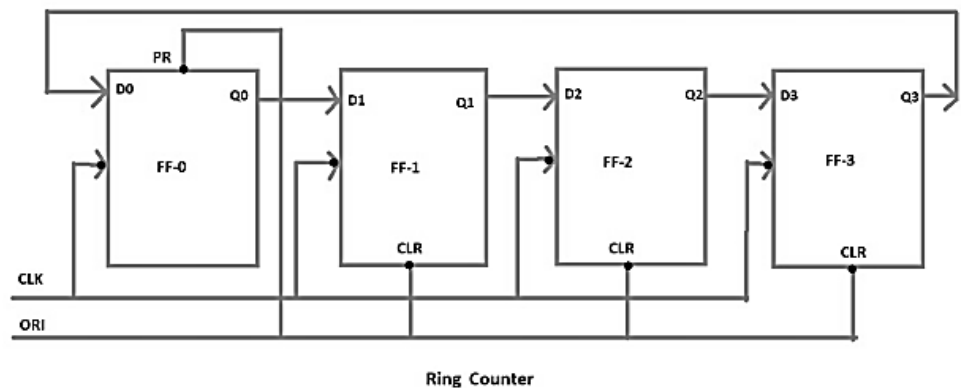
3. Clock Cycle 2: state 0010.

4. Clock Cycle 3: state 0001.

5. Clock Cycle 4: state 1000.

State Sequence:

$1000 \rightarrow 0100 \rightarrow 0010 \rightarrow 0001 \rightarrow 1000$.



Working Process:

- A Ring Counter is typically built using flip-flops (D, T, or JK flip-flops)
- A Ring Counter uses **n flip-flops**, where one flip-flop is set to **1** and the others are set to **0**.
- On each clock pulse, the **1 bit shifts** from one flip-flop to the next.
- The output of the **last flip-flop is fed back to the first**, creating a continuous repeating sequence.

20. Ring Counter Advantages and Disadvantages and Application.

Advantages:

- ❖ Simplicity
- ❖ Low Resource Use
- ❖ Predictable Output
- ❖ Easy to Decode
- ❖ Glitch-Free Output

Disadvantages:

- ❖ Limited States
- ❖ Not for Complex Patterns
- ❖ Needs Initialization
- ❖ Hardware Cost Increases

Applications:

- ❖ Sequence Generation
- ❖ Control Systems
- ❖ Simple Design
- ❖ Finite State Machines (FSM)
- ❖ Shift Register Based Applications

21. Explain n-bit Johnson Counter.

A Johnson counter, also known as a twisted ring or creeping counter, is a synchronous shift register counter. It uses n flipflops to generate $2n$ unique states.

Total number of used and unused states in n-bit Johnson counter:

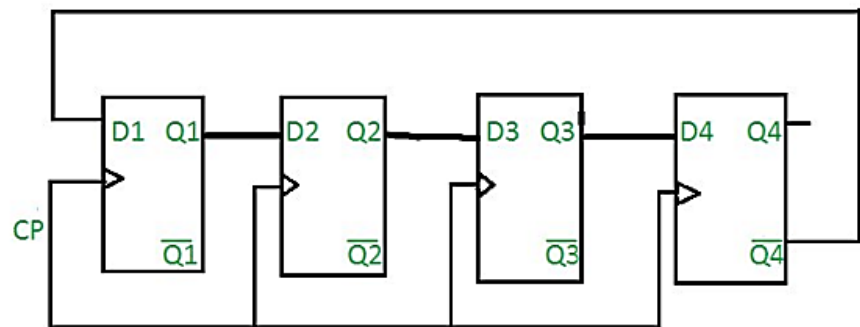
number of used states= $2n$

number of unused states= $2^n - 2*n$

Example:

If $n=4$

4-bit Johnson counter:



Truth Table

CP	Q1	Q2	Q3	Q4
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0

22. Johnson Counter Advantages and Disadvantages and Application.

Advantages:

- The Johnson counter has same number of flip flop
- It can be implemented using D and JK flip flop.
- Johnson counter is a self-decoding circuit

Disadvantages:

- Johnson counter doesn't count in a binary sequence.
- The number of flip flops needed is one half the number of timing signals.

Application:

- Johnson counter is used as a synchronous decade counter or divider circuit.
- It is used in hardware logic design
- The 3 stage Johnson counter is used as a 3 phase square wave generator
- It is used to divide the frequency of the clock signal

23. Ring Counter VS Johnson Counter.

Ring Counter	Johnson Counter
A ring counter employs the carry-in of the last flip-flop into the input of the first flipflop without any manipulation.	In Johnson counter, the complement of output of the last flip-flop is applied to the input of the first flip-flop.
n' flip-flops are required to count 'n' states.	'n' flip-flops are required to count '2n' states.
It can Generate 'n' unique states	It can Generate '2n' unique states
Unused states None, because all the states are utilized	'2n-2n' states are unused
Its not self-decoding since additional circuitry is needed	Its self-decoding makes it simpler for certain applications
Since it does not require inversion feedback	Due to inversion, the circuit is slightly more complex